GARUDA

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What is Garuda?

Garuda is an urban infrastructure technology initiative based on IoT (Internet of Things), designed specifically for the Barcelona of the future. Its main objective is to monitor, manage, and balance the city's avian ecosystem.

The system acts on three simultaneous fronts: real-time data collection, controlling the expansion of invasive exotic species, and actively promoting the recovery and reproduction of displaced native fauna.

Motivation and environmental context

In recent years, Barcelona has suffered a significant ecological impact due to exotic invasive species such as the monk parakeet (Myiopsitta monachus), whose population has increased by more than 5,000 individuals since 2015. This overpopulation poses a direct threat to local flora and fauna.

The monk parakeet actively displaces native species such as the house sparrow (Passer domesticus), competing aggressively for food sources and nesting spaces. Data shows that in areas with a high density of parakeet nests, the sparrow population is drastically reduced. Garuda's mission is to fight against this imbalance, protecting vulnerable local species from the territorial rise of invasive ones.

How does the smart birdhouse work?

The core of Garuda is a network of smart, automated birdhouses. Its purpose is to reduce the population of invasive species through an ethical, sustainable, and completely painless birth control method, allowing species like the sparrow to gradually recover their urban habitats.

To achieve this, the birdhouse features a dual-compartment design: one stocked with standard birdseed and another with birdseed treated with chemical castrators (administered exclusively during the mating seasons of invasive species).

Key Components and Functionalities

The system operates autonomously thanks to the integration of various sensors and algorithms:

- <u>Computer Vision and Access Management:</u> An integrated camera analyzes in real time what type of bird has landed on the birdhouse. Using mechanical actuators, the system opens or closes the doors of the food compartments. In this way, it guarantees that the modified birdseed is only consumed by the target invasive species, while the local fauna safely accesses the normal birdseed. Every interaction feeds an activity log and a population count.

- Acoustic monitoring: An environmental microphone captures and identifies the songs of nearby birds. This allows for the generation of a sound map and a detailed record of the species that frequent the ecosystem where the birdhouse is installed.
- Native Auditory Stimulation: An integrated speaker plays specific songs and calls designed to encourage the reproduction and safe settlement of sparrows and other local species that feel threatened by invasive ones.

Prototype architecture

Software

In the software architecture currently implemented in the prototype, the application developed for the Arduino Uno Q executes all functionalities and services locally. To implement these features, the Image Analysis module (Brick) has been integrated with our AI model, which was specially trained to differentiate between monk parakeets and sparrows. Furthermore, an Artificial Intelligence Audio Analysis module is used, which has been trained to distinguish between both species based on the songs detected by the webcam's microphone.

The website created for the Garuda Project displays the different locations of the birdhouses on a map; by selecting the fictitious location of the prototype, a real-time image from the installed camera is shown. Depending on what the AI model detects via the camera, the mechanical systems represented in the 3D model are triggered to open or close the corresponding doors. If a sparrow is detected, a sound simulating its species' song will be played.

The models are trained exclusively to detect birds. All images and audio are (and would continue to be) processed entirely locally to ensure privacy; therefore, if a person appears in the camera's field of view or a voice is captured, the model disregards it, as no data or metadata unrelated to birds is stored. Due to hardware availability limitations during this hackathon, a buzzer playing at different frequencies is used to simulate bird songs. All the AI models explained in this section have been developed using the Edge Impulse web application, putting into practice the knowledge acquired during the workshop sponsored by Qualcomm.

Hardware

The prototype's hardware architecture is based on an Arduino Uno Q board with 4 GB of RAM, equipped with the following Modulinos:

- 1 Buzzer.
- 2 RGB LED strips.
- 1 motion detector.

Additionally, a Logitech Brio105 webcam with FullHD quality (1080p) and an integrated microphone is used.

Possible future implementations

- Separate the architecture into independent servers: hosting the database and an API interface on one server, and the data visualization website on another.

- Make each Arduino Uno Q board responsible for collecting data and controlling access to bird food, sending the information via API calls.

- Optimize power consumption by making the camera turn on only when the presence detector senses motion.

- Replace the buzzer with a real speaker to play relevant bird songs instead of a simple synthesized melody.

- Expand the database so the models can detect and manage a wider variety of bird species.

- Use color codes on the LED strips to indicate the birdhouse's status, allowing for easy and visual identification of which birdhouses require maintenance (such as refilling the birdseed tank or fixing a malfunctioning component).

- Complement the visual warning by sending an automated error alert to the server so it is directly reflected on the web interface.

- Integrate the smart birdhouse alongside other products within an ecosystem of elements designed to make Barcelona a less hostile environment for native bird species. Proposed future add-ons include a smart car module (or an add-on for passenger vehicles) capable of detecting, recording, and analyzing city nests to create a comprehensive ornithological map, among other implementations to be determined (TBD).

Built with ❤️ for a more sustainable and balanced urban ecosystem.